

TECHNICAL SPECIFICATION AND METHODS

GSE42568 Methods Appendix

Frozen parameters, statistical models, gene universes and
reproducibility controls

Dataset: GEO GSE42568 · GPL570

Contrast: cancer minus normal breast tissue

Release: v1.0 · 11 September 2026

Scientific status: PASS WITH DOCUMENTED LIMITATIONS

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Scope and inputs

The formal contract is `GSE42568_PROJECT_DESIGN.md` : one cancer-minus-normal contrast with no WGCNA, survival, classifier, immune-infiltration or drug-prediction expansion. `scripts/download_inputs.sh` downloads the official matrix and GPL570 annotation snapshot. The matrix must match the frozen byte size, gzip test and SHA-256 before parsing.

The accepted matrix has 54,675 probe sets, 121 samples, range 2.3128–16.0469 and zero non-finite values. GEO identifies it as log2 GCRMA; no second log or normalization is performed.

Metadata and design

The series-matrix header supplies original and normalized metadata in `sample_manifest.tsv`. With no pairing ID and no normal-sample ages, the design is unpaired and does not adjust age. Cancer-only clinical covariates cannot be included in a joint cancer-normal model.

For gene g , $E[y_{gi}] = \text{beta_0g} + \text{processing_run_proxy_i} + \text{beta_group,g} I(\text{cancer_i})$. Normal is reference. The 121×14 design is rank 14, with 107 residual df, 13 run levels, six mixed levels and 62 mixed-run samples. `beta_group,g` is cancer-minus-normal log2FC.

Processed-data QC

QC includes sample median/IQR/MAD/quantiles/range, Pearson and Spearman connectivity, primary PCA and robust PCA distance. Warnings use robust $|z| > 3$. PCA uses top 5,000 pooled-MAD genes, centered but not scaled; top 1,000 and all-gene versions are sensitivity outputs. Robust distance uses up to 10 PCs reaching $\geq 50\%$ variance and a 0.999 contour. A sample reaches review only with at least two independent warning categories; no automated exclusion is allowed.

Annotation

Audit metric	Count/value
mapping_status:primary_eligible	39815
mapping_status:unmapped	9953
mapping_status:one_to_many	1524
mapping_status:constant	96
mapping_status:cross_hybridizing_x_at	3225
mapping_status:control_affx	62
suffix_class:s_at	11292
suffix_class:at	37394
suffix_class:a_at	1720
suffix_class:x_at	4207
suffix_class:control	62
total_probe_sets	54675
noncontrol_nonconstant_denominator	54517
primary_eligible_probes	39815
primary_eligible_fraction	0.730322651649944
unique_testable_entrez	20357
genes_with_multiple_eligible_probes	10281

Mapping uses `hgu133plus2.db 3.13.0` and `org.Hs.eg.db 3.22.0`. Stable keys are Entrez IDs; symbols are display snapshots. AFFX controls, constants, unmapped, one-to-many and `_x_at` probes are excluded. The representative order is `_at`, `_a_at`, `_s_at`; ties use pooled median then probe ID. Group labels and statistics are never used.

Median collapse and all eligible probe fits provide independent summarization/technical layers. A material opposite probe is opposite in direction and itself significant by probe-level 1.2-fold TREAT BH FDR.

DE and sensitivity

P1 uses limma `lmFit`, `eBayes(trend=TRUE, robust=TRUE)`, cancer-minus-normal coefficient, BH across all 20,357 genes, and posterior-SE 95% CIs. Discovery is moderated BH FDR <0.05. Priority is `treat(lfc=log2(1.2))` BH FDR <0.05.

Minimum FC	Minimum log2FC	TREAT FDR <0.05	Up	Down
1.1	0.138	3730	2027	1703
1.2	0.263	2835	1497	1338
1.5	0.585	1451	680	771

- S1: group only, all 121 samples.
- S2: run + group in six mixed runs (n=62).
- S3: not applicable; no signed technical failures.
- QC influence: remove four reviewed candidates (n=117).
- S4a: `trend=FALSE` ; S4b: group-aware `voomaByGroup` weights.
- Median collapse, probe-level and 13 leave-one-run-out fits complete the locked sensitivities.

Comparison	n	Spearman	Pearson	Priority sign	Median effect ratio
S1_group_only	121	0.936	0.975	100.0%	1.069
S2_mixed_runs	62	1.000	1.000	100.0%	1.000
QC_candidate_exclusion	117	0.912	0.975	100.0%	1.100
S4_no_trend	121	1.000	1.000	100.0%	1.000
S4_group_aware	121	0.962	0.993	100.0%	1.006
median_collapse	121	0.874	0.902	97.0%	0.960

Green requires Spearman ≥ 0.90 and priority-sign concordance $\geq 95\%$ for both P1/S1 and P1/S2; Amber minima are 0.80. Final status is Green. Headline genes must retain direction across core models, agree with median collapse and have no material opposite probe.

Visualization

F01–F15 are vector PDF/SVG and 600-dpi PNG with caption and source TSV in the complete analysis workspace. Adjusted PCA is visualization-only. Volcano categories use BH FDR despite a raw-P y-axis. Heatmap selection is TREAT-first, ≤ 25 genes per direction; row z-scores cap at ± 2.5 and never enter DE.

Enrichment and licensing

The universe is P1-tested unique Entrez genes annotated in the relevant resource. GO BP/CC/MF ORA tests cancer-up and normal-up separately, one-sided hypergeometric, post-intersection size 10–500, BH within ontology $\times$ direction. Reactome ORA uses the same rule. Reactome GSEA ranks all P1 genes by moderated t descending, stable Entrez tie-break, exponent 1, size 15–500 and `fgseaMultilevel` ; observed exact ties are zero.

Collection	Status	Version	Universe	Eligible sets	Ranking
GO_BP_ORA	completed	org.Hs.eg.db 3.22.0; GO.db 3.22.0	15588	6187	NA
GO_CC_ORA	completed	org.Hs.eg.db 3.22.0; GO.db 3.22.0	16428	756	NA
GO_MF_ORA	completed	org.Hs.eg.db 3.22.0; GO.db 3.22.0	16139	1233	NA
Reactome_ORA	completed	1.95.0	10178	1658	NA
Reactome_preranked_GSEA_fallback	completed	1.95.0	10178	1367	P1 moderated t, descending; exact ties by Entrez ID
KEGG_ORA	not_run_license_gate	NA	NA	NA	NA
MSigDB_Hallmark_GSEA	not_run_license_review_unconfirmed	NA	NA	NA	NA

Runtime and reproducibility

The runtime is defined by the pinned Conda records in `config/`; commands resolve from the active `PATH` or an optional `BIOINFO_ENV_PREFIX`. The verified environment used R 4.5.3, limma 3.66.0, fgsea 1.36.2, hgu133plus2.db 3.13.0, org.Hs.eg.db 3.22.0, GO.db 3.22.0 and reactome.db 1.95.0. `tests/run_tests.R` contains 13 guards; `R/05_scientific_qa.R` contains 32 scientific acceptance checks. A clean rebuild reproduced 10/10 key machine-readable outputs with identical SHA-256 hashes.

Methods references: [limma](#), [hgu133plus2.db](#), [GO](#), [Reactome](#), [fgsea](#).